

Request for Information: Identification of Potential Shore Installation Contractor Owned/Operated Nuclear Power Sites

07 October 2024

This is not a Request for Proposals (RFP).

The Department of the Navy (DON), which comprises the United States Navy and the United States Marine Corps, is seeking information and market interests from developers, utilities, consortia, or similar private or public parties regarding interest in developing non-excess, under-utilized DON lands for the purpose of constructing commercial scale, grid facing clean energy generation to achieve full base energy resilience. This initiative underscores the DON's commitment to enhancing resilience, energy efficiency, and decarbonization across the globe. This RFI aims to identify industry capabilities as well as solutions that mutually solve the DON's resiliency and carbon free energy goals, as well as improving the cost, timeline, and risk profile of industry projects of mutual interest with the DON. We welcome input from developers, utilities, and consortia on innovative approaches that may include, but are not limited to, Shore Based Nuclear Technologies and other advanced technologies. Our goal is to enhance energy security and reliability for DON installations and the surrounding communities, and we invite comments on various technologies, business models, contracting methods, and partnerships that can contribute to resilient, redundant energy systems in the face of evolving geopolitical challenges, climate change, and emerging threats across the DON.

BACKGROUND:

Installation resilience is a top priority for the DON, with a focus on achieving full resilience across the enterprise to ensure operational continuity. This is crucial in maintaining national security and readiness in the face of potential disruptions to commercial energy networks, whether from physical or cyberattacks, extreme weather, pandemic biothreats, or other emerging challenges. Ensuring that DON installations can continue their mission in the event of a grid power outage is a critical priority. The DON also recognizes that our installations, operations, workforce and supply chains are supported by critical infrastructure and community services that are privately owned and operated, off-installation. The DON is taking proactive steps to achieve a more sustainable and secure energy future while fulfilling its critical mission requirements.

The increasing frequency and severity of extreme weather events, the significant rise in cyber events, the potential for physical attacks as well as routine power outages necessitate a proactive approach to enhancing the resiliency of DON shore installations worldwide. In support of this approach, the Navy is committed to exploring the latest technological advancements, which offer the potential for robust solutions to these challenges. Technologies such as (but not limited to), advanced, distributed-scale shore based nuclear technology or other advanced reactor technology, when sited on or adjacent to DON installations, could provide stable and reliable

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energy for extended periods, ensuring that Navy and Marine Corps installations remain operational even in adverse conditions. Shore based nuclear technologies are becoming more attractive potential options, offering scalable and flexible solutions that can be tailored to the specific energy needs of a region while providing resilience needs of a DON military installation.

Implementations of new energy generation technology may be particularly attractive in serving large community and user bases, such as those proximate to many DON installations, while encumbering a relatively small land area. Under normal circumstances, the commercial electric transmission and distribution grids would be the primary off-taker from any generating station hosted on DON lands; in the event of grid disturbance, power would revert back first to the installation.

To realize the potential of new energy generation technologies for installation resilience, the DON is interested in exploring various technologies, business models, contracting methods, policies, and partnerships that would enable contractor-owned and operated power sites on DON installations, ensuring that the DON can leverage commercial expertise and innovation.

Partnerships with industry leaders, academic institutions, and other government agencies will be crucial in advancing the research, development, and deployment of technologies at DON installations. Through these collaborative efforts, the DON can achieve enhanced resilience, ensuring that its installations are capable of sustaining operations under any circumstances, thus reinforcing national security and operational readiness.

SCOPE OF WORK:

The DON is committed to improving installation and community resilience in accordance with its authorities from Title 10, U.S.C. 2911 and 2920. This RFI underscores the DON's commitment to align with national goals, aiming to maximize community and installation resilience and achieve statutory requirements for 99.9% energy reliability for hosting installations supplied through the commercial distribution systems as well as assuring a minimum of 14 days energy endurance during a commercial distribution grid outage. We will also seek to achieve net-zero greenhouse gas emissions by 2050. Additionally, the DON is dedicated to supporting the National commitment to triple nuclear power generation by 2050. As articulated by the Honorable Meredith Berger, Assistant Secretary of the Navy for Energy, Installations, and Environment (ASN EI&E), climate change amplifies threats to the Navy's personnel, installations, platforms, operations, and international alliances. To uphold its status as the preeminent maritime force worldwide, the DON underscores the critical necessity to fortify resilience and diminish environmental vulnerabilities. In the face of escalating climate change impacts, the DON recognizes the imperative to enhance its operational capabilities while simultaneously mitigating environmental risks.

One of the avenues the DON is exploring to increase installation resilience, reduce carbon

emissions, and enhance operational capabilities is the potential for privately-owned and operated clean energy power generation stations on DON installations. This Request for Information (RFI) aims to gather insights, expertise, innovative business models, effective contracting methods, robust policies, and potential partnerships from industry experts, developers, and stakeholders. The focus is on four (4) prioritized Navy installations and/or three (3) USMC installations identified for initial implementation (see below). The DON is open to exploring additional installation(s) which could be used to for contractor owned/operated power generation site(s). While seven (7) installations from both the Navy and USMC have been identified as the priority, respondents should state their preference of installation(s) and considerations that will need to be made for each installation selected, as well as for any other DON installations that respondents may identify for consideration.

The DON welcomes innovative solutions, technologies, and concepts which can be utilized to achieve reliable power quality, black start capability, independent microgrid capability, continuous safe operations and distribution, and meet peak critical load requirements 24 hours a day during a prolonged disruption in commercial energy supplies, thereby enhancing energy security, operational capability, strategic flexibility and resource availability for DON installations. Contractors who own or operate nuclear power sites on Navy or USMC installations would be responsible for the possession, storage, and management of nuclear and spent fuel, as well as obtaining and complying with Nuclear Regulatory Commission (NRC) licensing requirements. The DON will be looking at developer responses for creative solutions for the storage and disposal of spent fuel in accordance with NRC requirements.

DON Strategic Objectives Associated with this RFI

Identify Opportunities that Enhance and Support Installation and Community Energy

Resilience: Collect input for privately-funded concepts that would ensure continuous operational capability of DON installations in the event of grid power outages, adversary attacks, extreme weather events, or other disruptions, while improving utility regional grid capacity, flexibility, affordability and resilience to support community continuity of services.

Identify Opportunities that Enable and Advance the Safe, Secure, and Responsible Development of Domestic, Distributed-Scale, Advanced Nuclear Power Reactor and Fuel

Technologies: Collect input for privately-funded concepts that would advance, accelerate and expand U.S. commercial markets, domestic supply chains, workforce development, industry capacity, technology diversification, and economic viability for advanced, safe, secure, distributed-scale, NRC-licensed nuclear power reactor and fuel technologies, that DON installations can support and further enable.

Identify Opportunities and Challenges Resulting from Policy, Regulatory and Economic Issues that Could Enable or Hinder U.S. Domestic Development of Safe, Secure, and Responsible Advanced, Distributed-Scale Nuclear Power and Fuel Production for Commercial Markets. Collect input regarding opportunities and challenges presented by policy,

regulatory or economic issues that impact the ability of DON installations for hosting privately-funded advanced, distributed-scale nuclear power plant concepts that could serve regional grid power needs while also providing improved energy resilience for DON mission operational power requirements.

Leverage Industry Expertise: Utilize the knowledge and experience of industry experts to inform the DON's strategy for nuclear energy deployment.

Strengthen Benefits to the Wider Community: Collect input regarding concepts that would provide power to the community rate base as the primary power customer, improving power quality, reliability and resilience for the area populace, fostering community industrial and commercial development, tax base growth and jobs growth.

Candidate Installation

The DON has identified that the following installations have the potential to be mission-compatible candidates seeking full base resilience.

Virginia

- Naval Air Station (NAS) Oceana (VA)
- Naval Support Activity (NSA) South Potomac (includes Naval Support Facility (NSF) Dahlgren, VA and NSF Indian Head, MD)
- Naval Weapons Station (NWS) Yorktown (VA)
- Marine Corps Base (MCB) Quantico (VA)

Maryland

- NAS Patuxent River (MD)

North Carolina

- Marine Corps Air Station (MCAS) Cherry Point (NC)
- MCB Camp Lejeune (NC)

RFI Response Requirements

1. Executive Summary - Provide a brief overview of your organization and its experience with nuclear power solutions. Summarize your insights on the feasibility and benefits of contractor-owned and operated nuclear power sites for DON installations.

2. Technical Expertise and Capabilities - Detail your technical expertise and capabilities in the design, construction, operation, and maintenance of the proposed technology. Highlight any previous projects or case studies relevant to military or critical infrastructure energy resilience.

3. Business Models - Propose innovative business models for contractor-owned and operated clean energy power sites. Explain how these models can ensure cost-effectiveness, reliability, and long-term sustainability. Address financing mechanisms, risk-sharing, and revenue models.

4. Contracting Methods - Suggest effective contracting methods that could facilitate the deployment of clean power generation at Navy and Marine Corps installations. Include recommendations on contract structures, performance incentives, risk mitigation strategies, and lifecycle management.

5. Policy and Regulatory Considerations - Identify key policy and regulatory challenges and propose solutions to address them. Discuss safety, security, and environmental compliance requirements. Provide insights on the regulatory approval process and how it can be streamlined.

6. Partnerships and Collaboration - Propose potential partnerships and collaboration frameworks with industry, government agencies, and academic institutions. Highlight how these partnerships can enhance innovation, reduce costs, and accelerate deployment.

7. Implementation Plan - Outline a high-level implementation plan for deploying clean energy generation at one of, or multiple, prioritized DON installations. Include timelines, key milestones, and resource requirements (acres, locality of water and/or distribution power lines, etc.) and additional considerations required to continue providing power to Navy and Marine Corps installations from the nuclear power asset in the event of a grid outage.

8. Risk Management - Identify potential risks associated with the deployment and operation of power sites and propose mitigation strategies. Include risks related to technology, regulation, finance, climate change, sea level rise and security.

RFI Response Submission

Responses to this RFI should be submitted no later than **07 November 2024**.

Submit responses to this RFI to: NAVFACHQ-RE@us.navy.mil

Please use the subject line “Response to **Installation Energy Resilience; identification of potential technology solutions** RFI”. No limit on response length but please limit file sizes to 5MB.

This RFI is for informational purposes only and does not constitute a solicitation for proposals or offer of contract. The information gathered will be used to inform the Department of the Navy’s strategy for enhancing installation resilience through innovative and new energy solutions.